US012414872B2

(12) **United States Patent**
Lyndon

(10) **Patent No.: US 12,414,872 B2**
(45) **Date of Patent: Sep. 16, 2025**

(54) **DEVICE FOR SUPPORTING A LOWER LIMB**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 366 days.

(21) Appl. No.: **17/912,210**

(22) PCT Filed: **Mar. 29, 2021**

(86) PCT No.: **PCT/AU2021/050278**
§ 371 (c)(1),
(2) Date: **Sep. 16, 2022**

(87) PCT Pub. No.: **WO2021/195696**
PCT Pub. Date: **Oct. 7, 2021**

(65) **Prior Publication Data**

US 2023/0134809 A1 May 4, 2023

(30) **Foreign Application Priority Data**

Apr. 3, 2020 (AU) 2020901035

(51) **Int. Cl.**
A61F 5/058 (2006.01)

(52) **U.S. Cl.**
CPC ***A61F 5/0585*** (2013.01); *A61H 2201/0192* (2013.01); *A61H 2201/12* (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC A61H 2003/005; A61H 2201/0192; A61H 2201/12; A61H 2201/1652;
(Continued)

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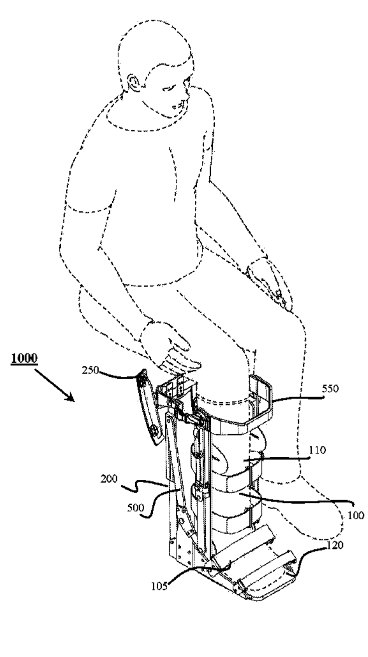
(57) **ABSTRACT**

A device for supporting a lower limb of a subject. The device has a housing for a lower limb and a limb support. The housing has a longitudinal axis aligned with a longitudinal axis of the lower limb. A driving mechanism connected between the housing and the limb support, causes movement of the limb support between a retracted position parallel to the longitudinal axis of the housing and an extended position in which the limb support extends perpendicularly to the longitudinal axis of the housing to support the lower limb.

19 Claims, 11 Drawing Sheets



(52) **U.S. Cl.**
CPC *A61H 2201/1652* (2013.01); *A61H 2203/0437* (2013.01)

(58) **Field of Classification Search**
CPC A61H 2201/1642; A61H 2201/1253; A61H 2201/165; A61H 2203/0437; A61H 3/00; A61F 5/0585; A61F 5/3761; A61F 5/0123; A61F 5/0127; A61F 5/013; A61F 5/37; A61F 5/01; A61F 5/0102; A61F 5/0125; A61F 5/0167; A61F 5/0158; A61F 5/02; A61F 5/30; A61F 5/0195; A61F 5/00; A61F 2005/0137; A61F 2005/0165; A61G 7/075; A61G 7/0755; A47C 7/50; A47C 16/00; A47C 16/02; A47C 16/025
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See application file for complete search history.

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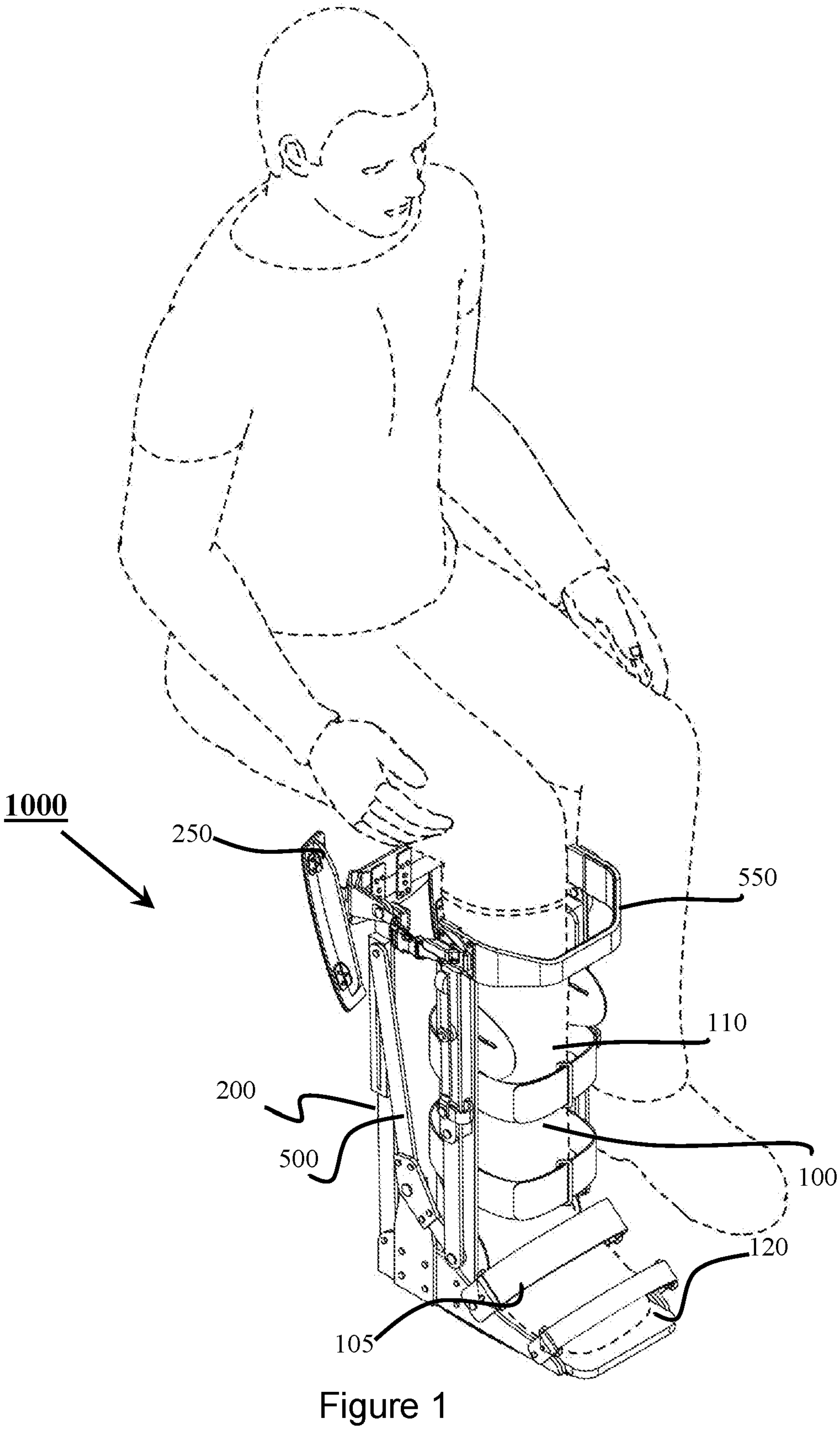
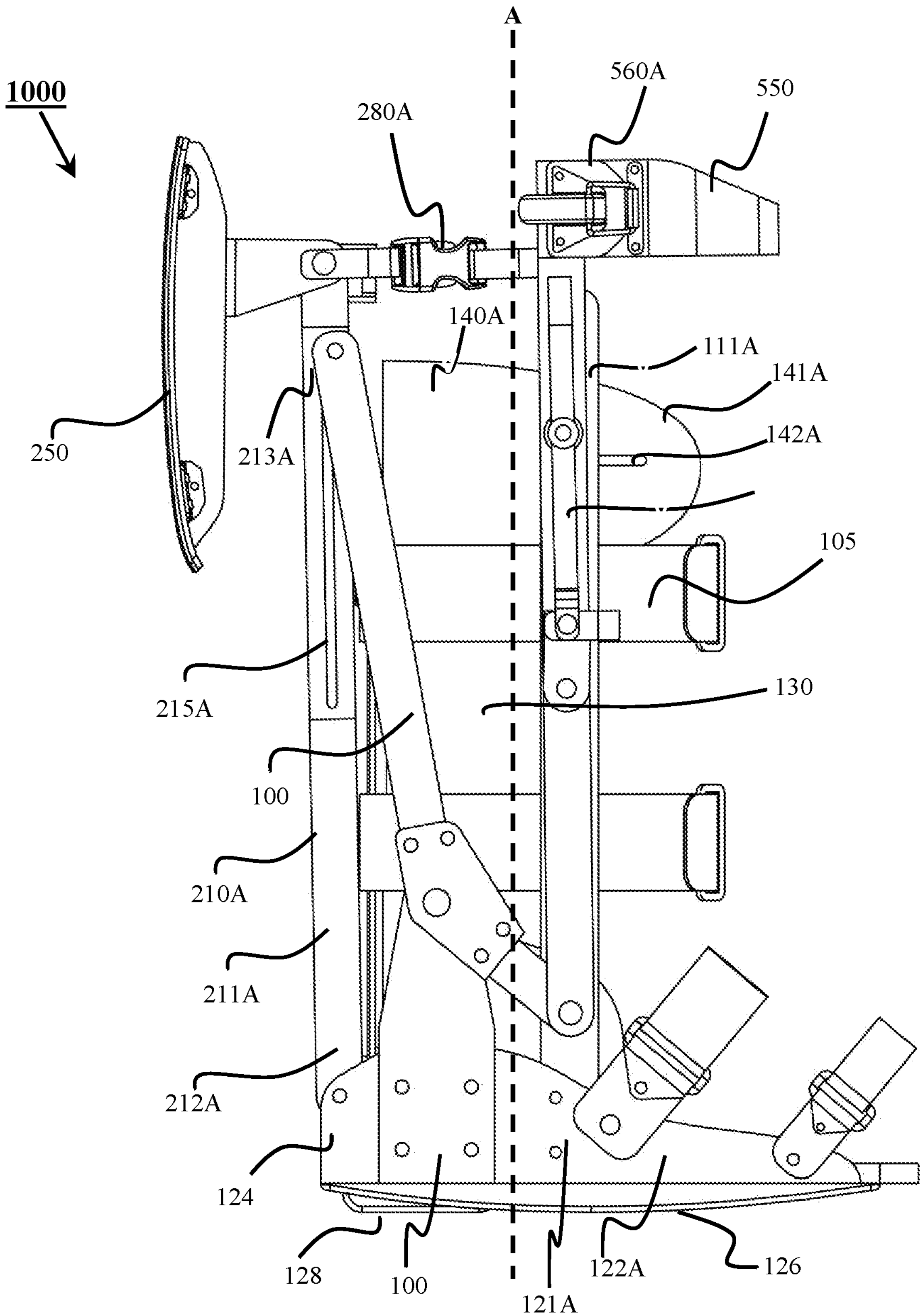


Figure 1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1000
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550
140A
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141A
142A
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121A
122A
126

Figure 2

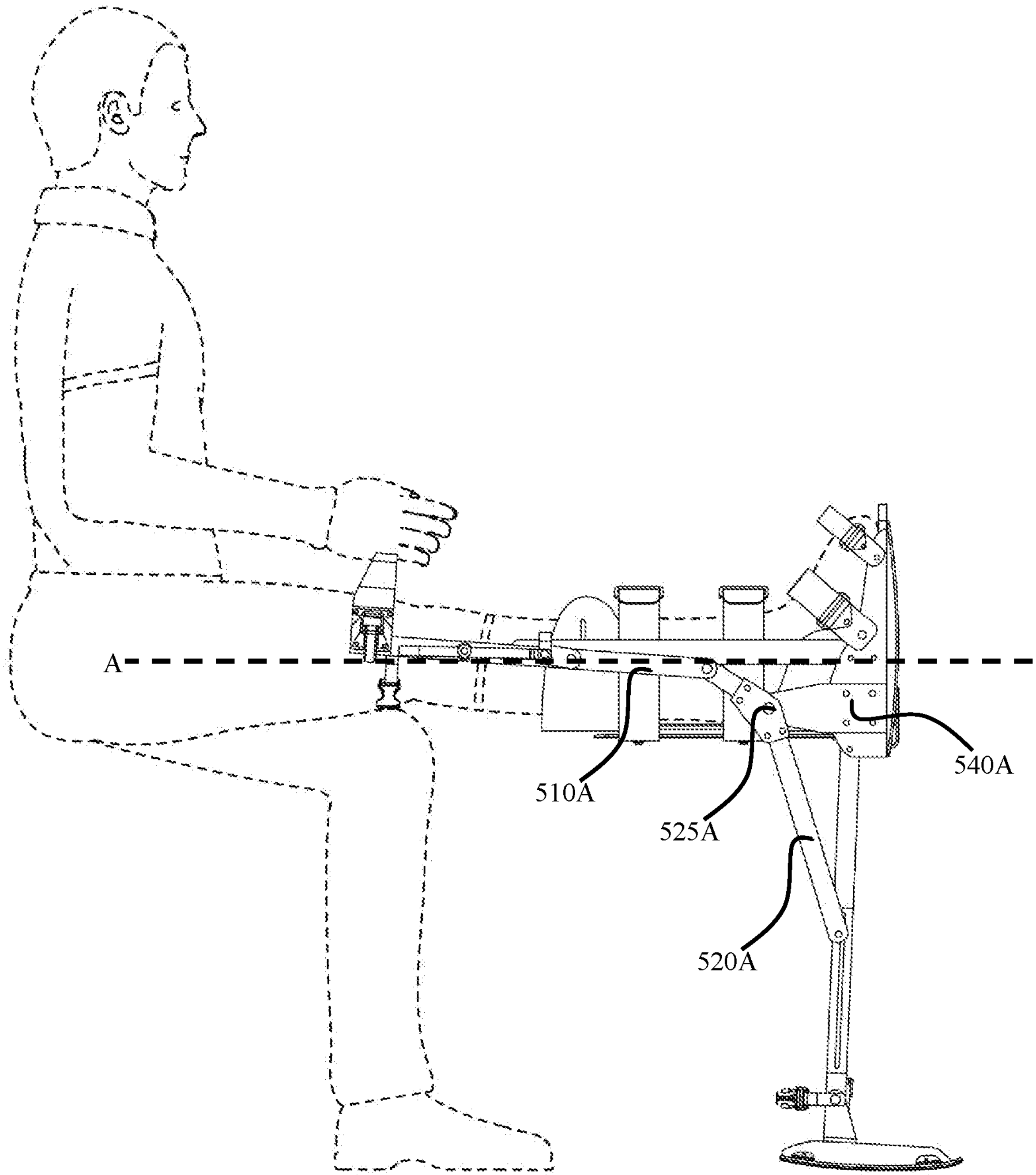


Figure 3

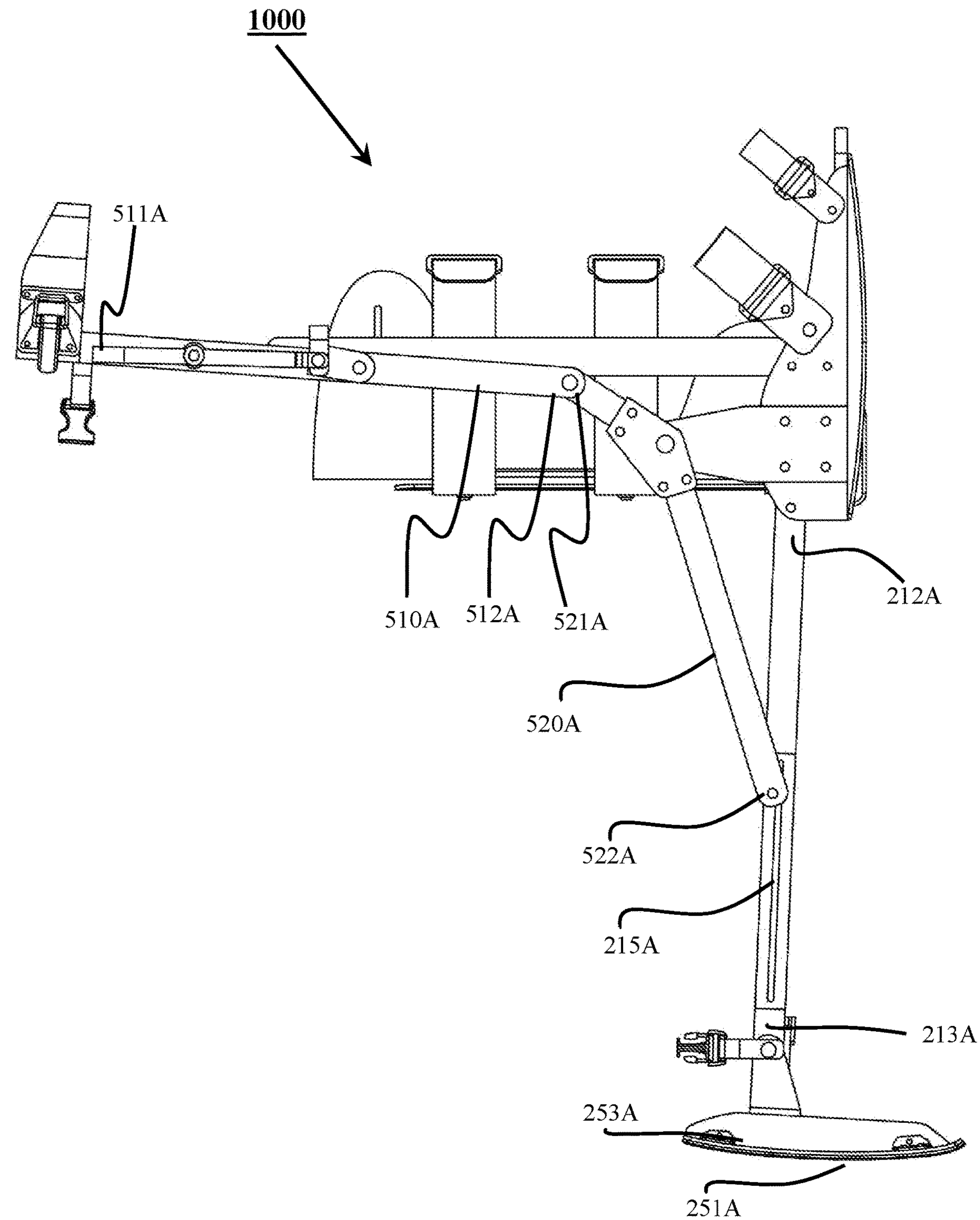


Figure 4

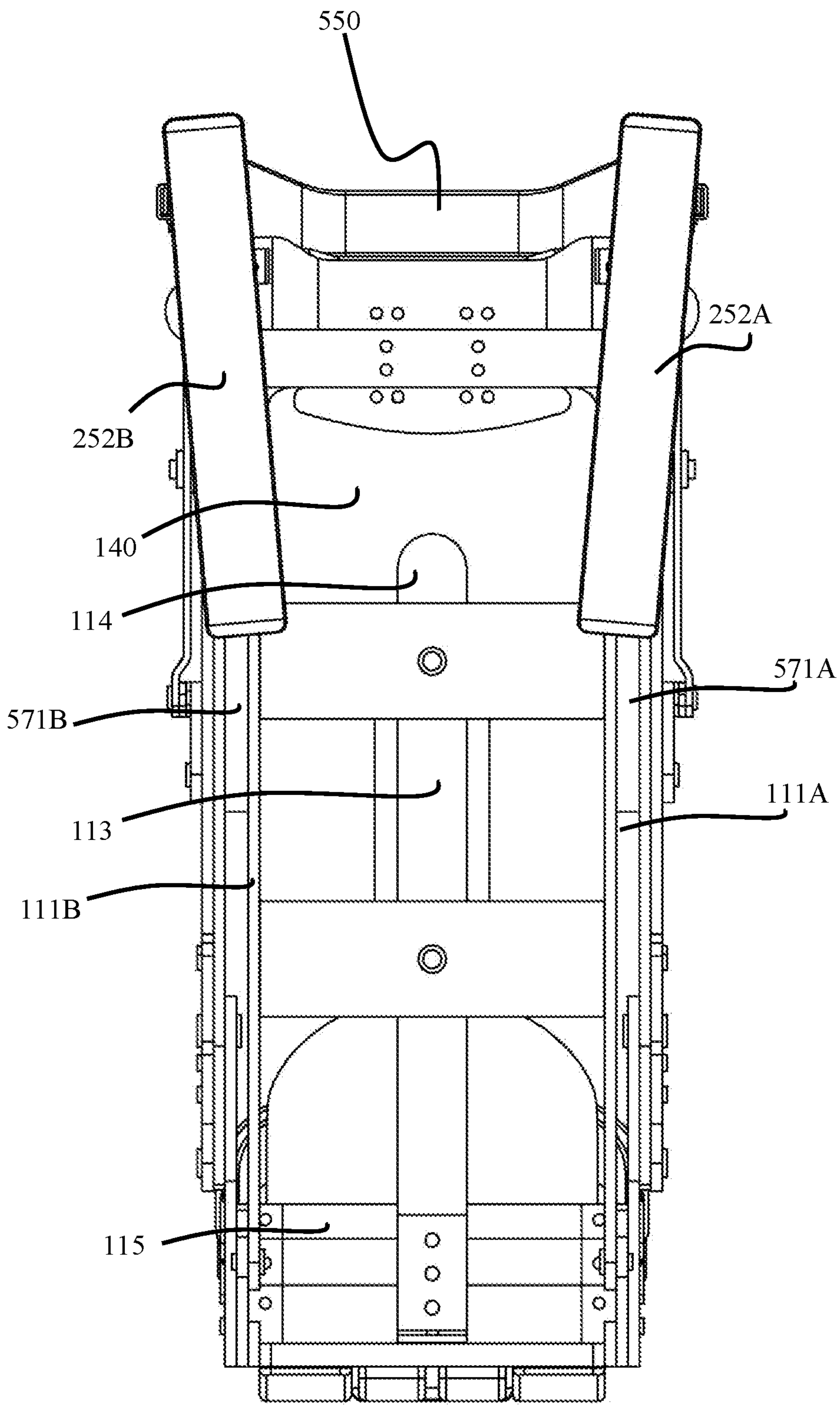
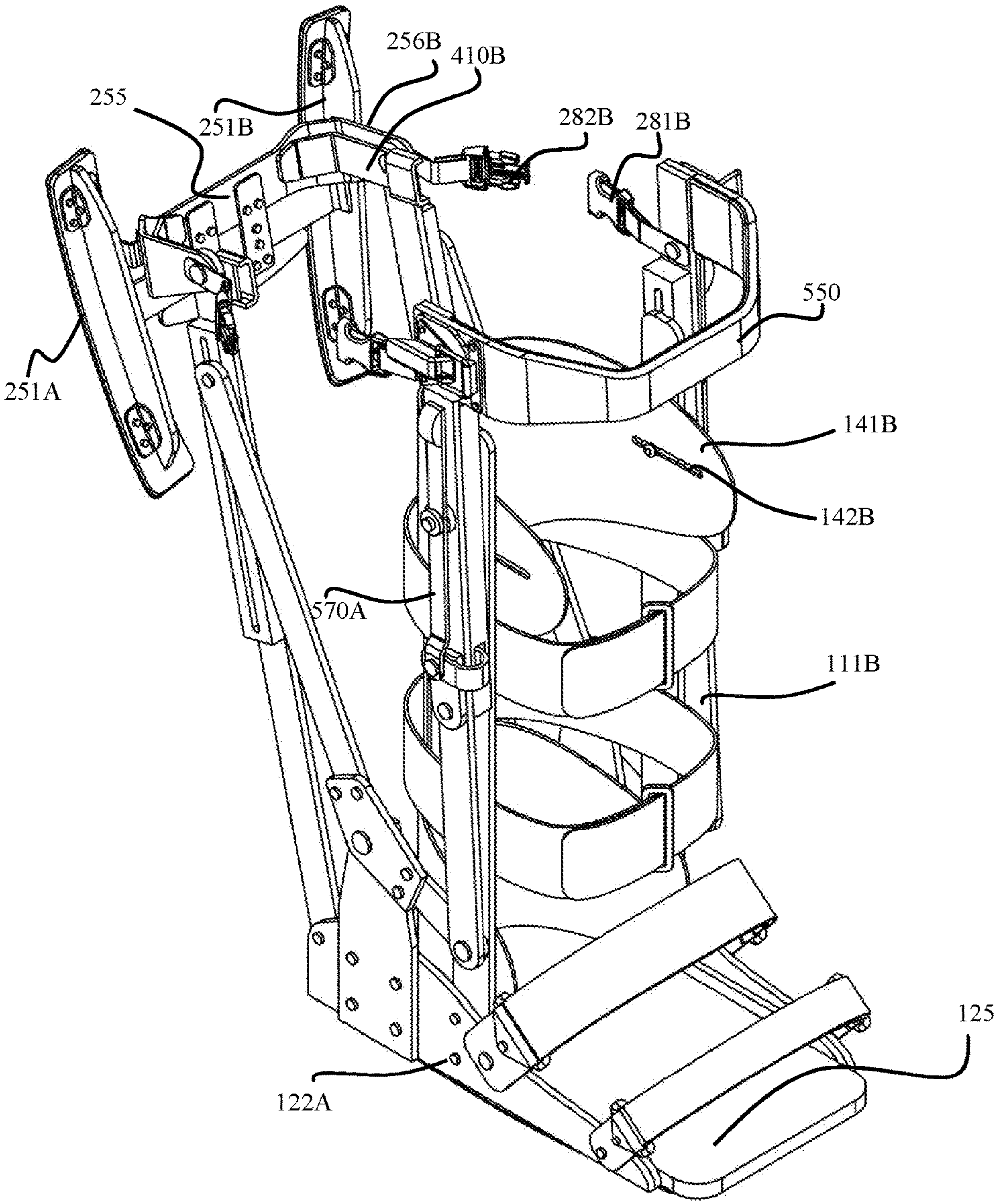


Figure 5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255
251B
282B
281B
550
251A
141B
142B
570A
111B
125
122A

Figure 6

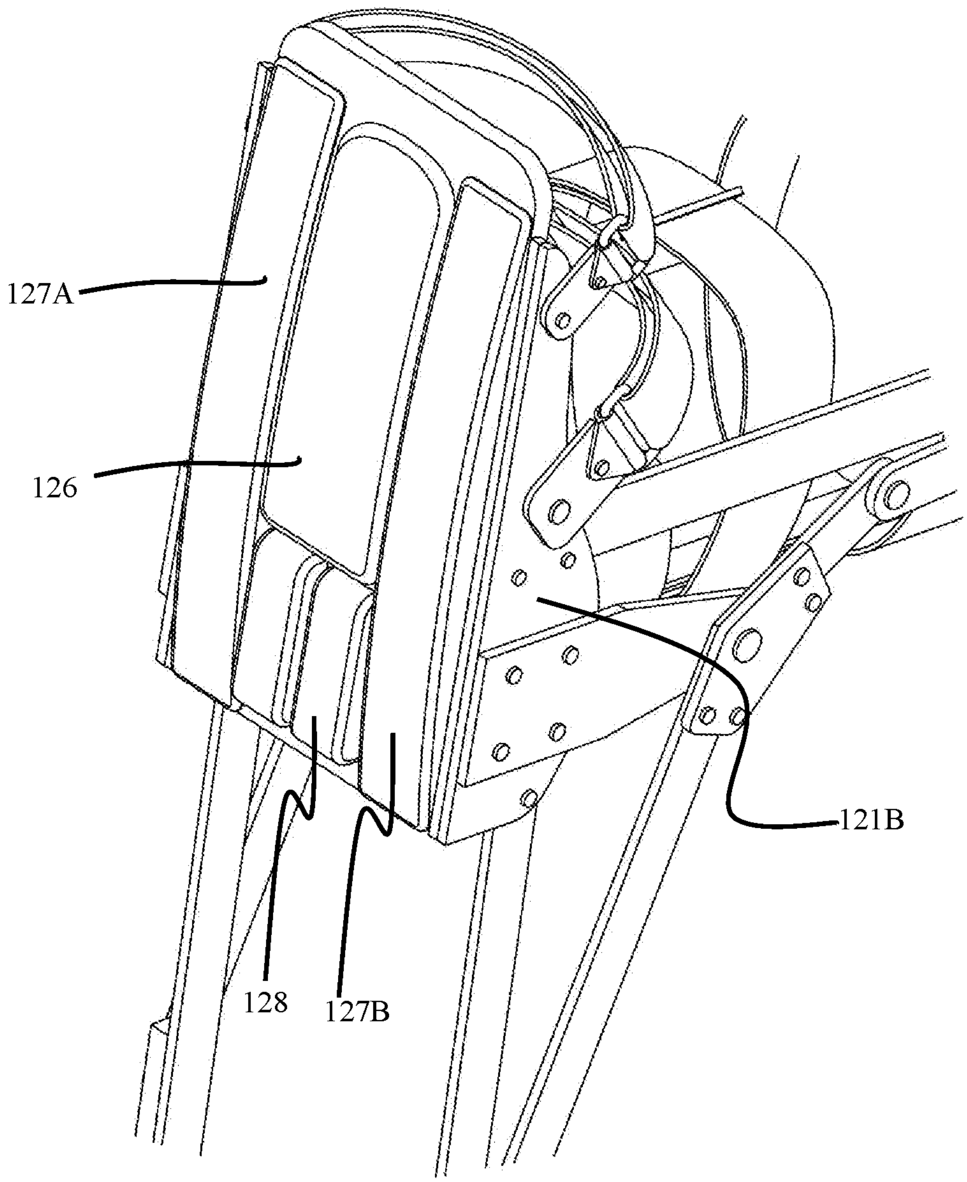

Figure 7

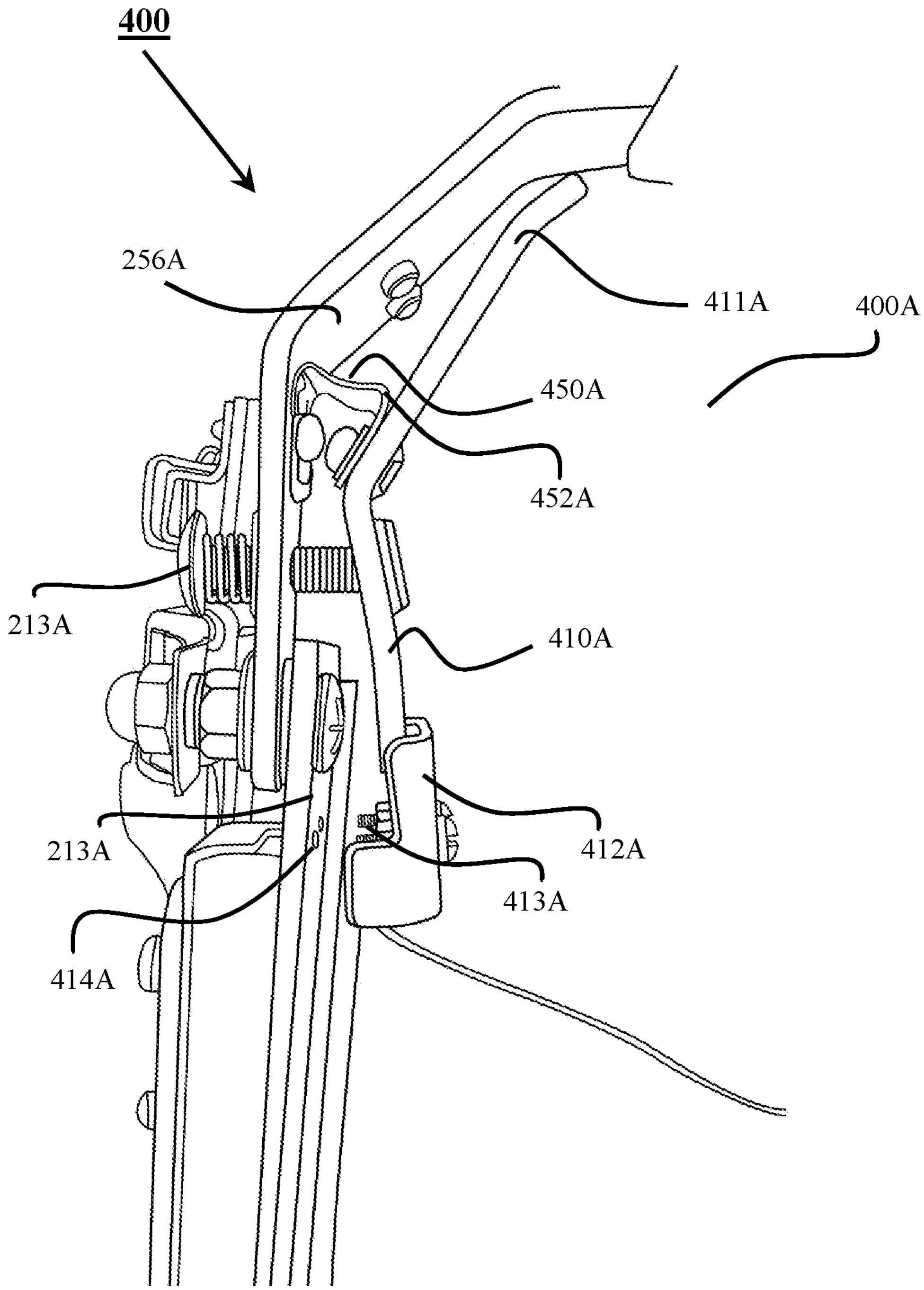

Figure 8

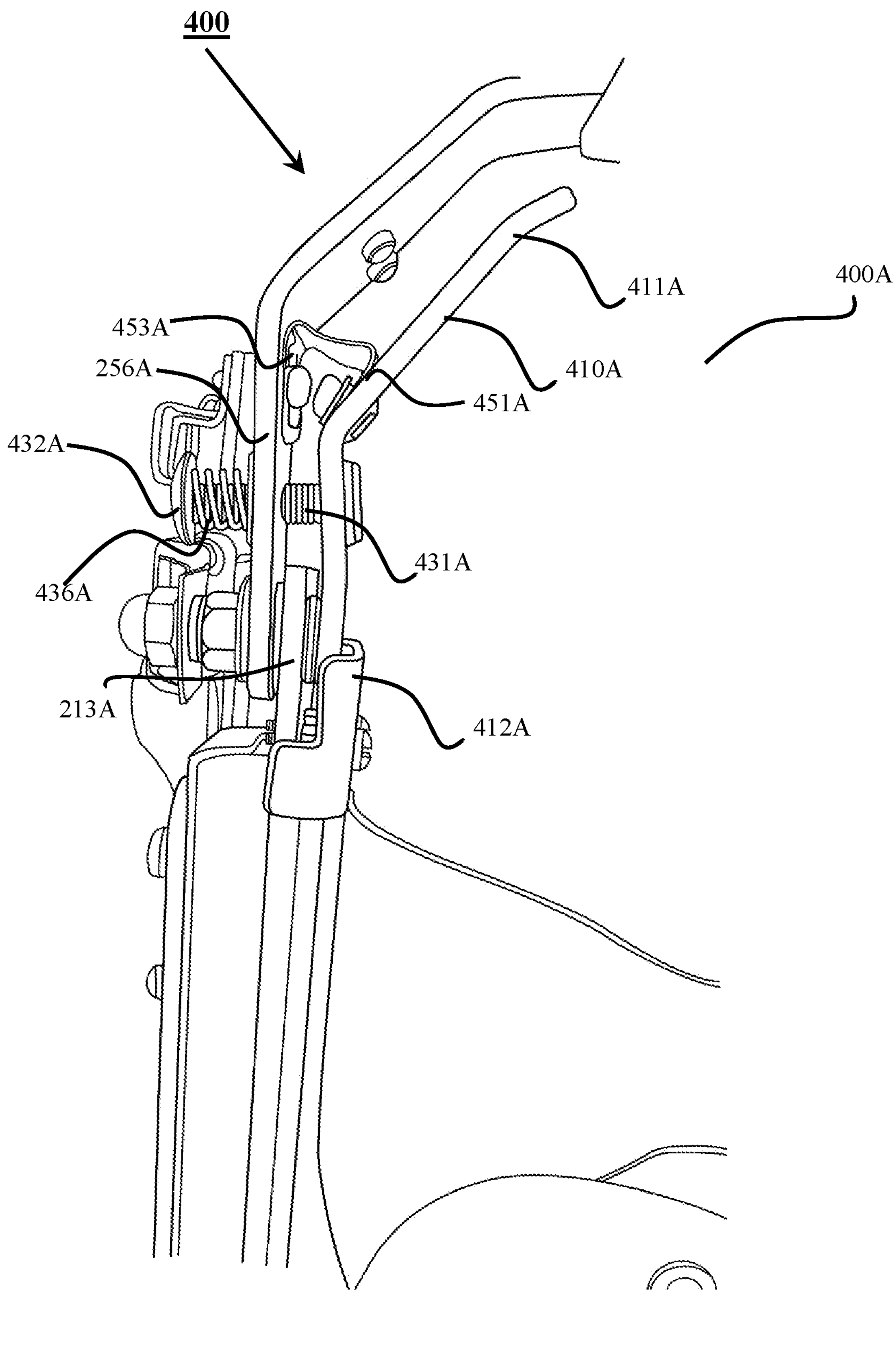

Figure 9

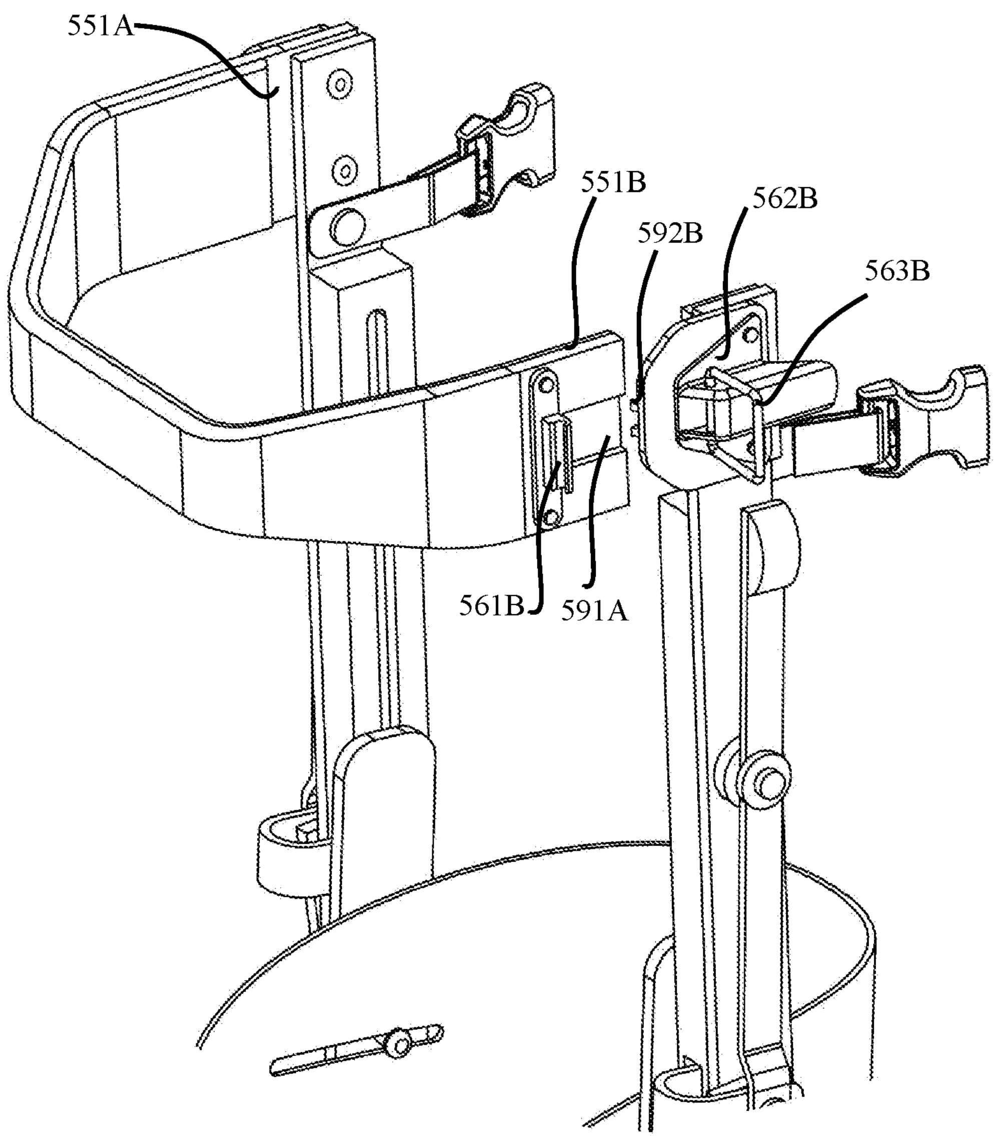

Figure 10

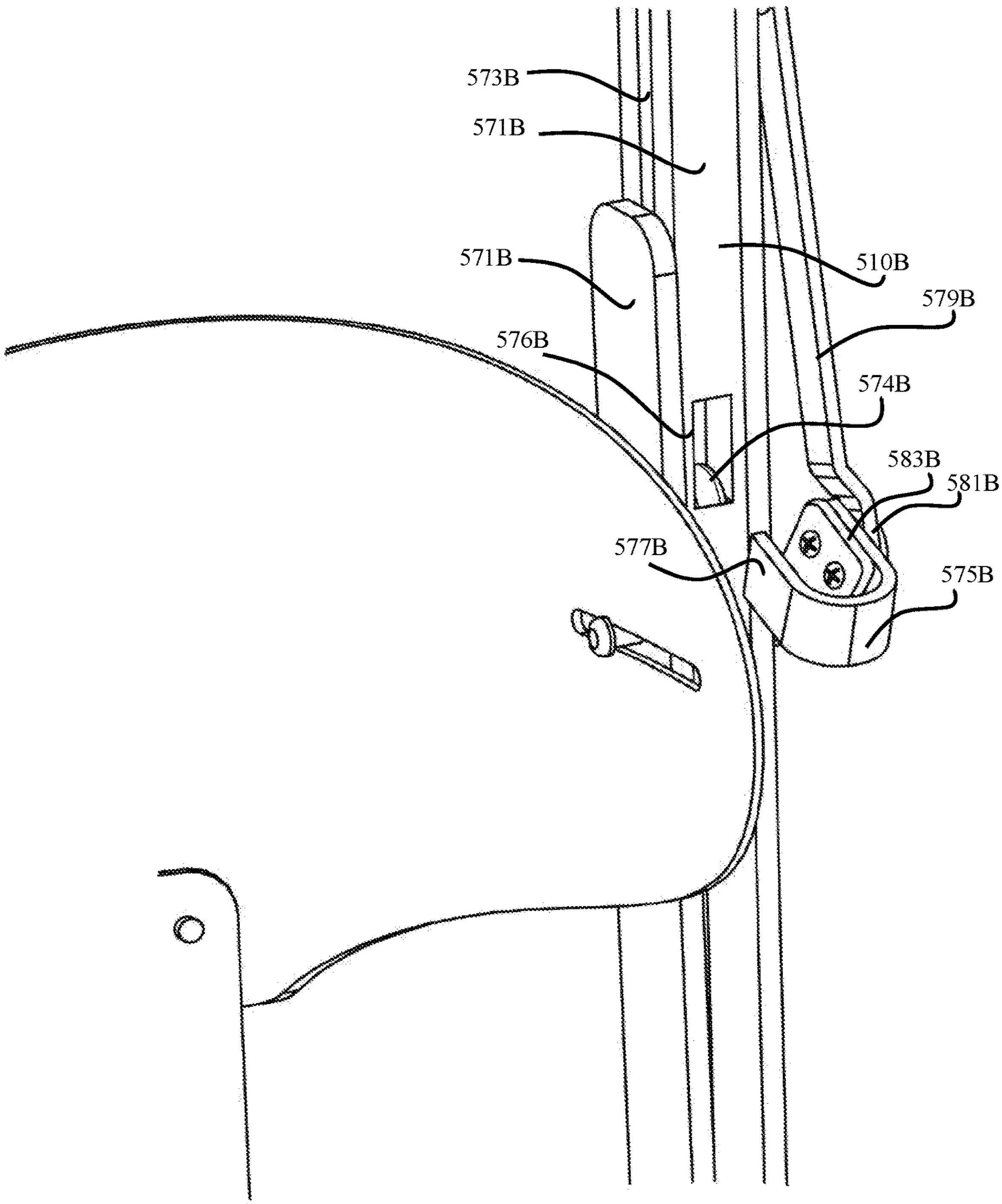

Figure 11

DEVICE FOR SUPPORTING A LOWER LIMB

FIELD OF THE INVENTION

The present invention relates to device for supporting a lower limb and in particular, to a device which can support a lower limb of a patient in an extended position of the lower limb, at a position elevated from the ground when the patient is seated.

The invention has been developed primarily for use while a patient is recuperating from an injury to a lower leg such as a fracture or surgery or other trauma and will be described hereinafter with reference to this application. However, it will be appreciated that the invention is not limited to this particular field of use.

BACKGROUND OF THE INVENTION

Typically, when a patient is recovering from surgery or other trauma such as a fracture in a lower leg, they are advised by a physician to elevate the injured lower leg to promote venous blood flow. Return of blood from lower extremities is inhibited by swelling of the lower limb. This is caused by interruptions to the vasculature by the trauma and the immune response to this trauma. There is typically a higher osmotic pressure in the tissues surrounding the veins than in the veins themselves which causes fluid to be drawn into the surrounding tissues.

The lower limb needs to be elevated relative to the heart of the patient to harness the force of gravity to assist venous return. When the patient is confined to bed, various supports can be used to elevate the lower limb. For example, the lower limb may be tired to a frame using one or more ties.

When a patient no longer requires bed rest, they are usually prescribed to wear a protective orthotic boot to protect the injured lower leg. The orthotic boot cushions the lower leg while providing mechanical stability to the lower limb.

Typically, after the patient is discharged from the hospital, they are still required to wear the orthotic boot for an extended period of time and also, to elevate their lower leg during the day, while recuperating.

The patient may use a stool or other type of furniture to rest their foot on while sitting. However, the stool must be the correct height to effectively elevate the limb and also, be comfortable to the user.

When the patient is not at home and a stool or other furniture at the right height is not available, the patient does not have a convenient option to elevate their limb.

If it is not convenient to elevate their lower limb while at home or outdoors, then the patient may not be motivated to stick to the prescribed therapy of elevating their lower limb. This may cause the patient to develop a significant amount of oedema in their lower limb and as a result, may take longer to recuperate and/or result in other medical problems.

The present invention seeks to provide a solution which will overcome or substantially ameliorate at least some of the deficiencies of the prior art, or to at least provide an alternative.

It is to be understood that, if any prior art information is referred to herein, such reference does not constitute an admission that the information forms part of the common general knowledge in the art, in Australia or any other country.

SUMMARY OF THE INVENTION

In an aspect of the present disclosure, there is provided a device for supporting a lower limb, comprising:

- a housing for a lower limb, the housing having a longitudinal axis;
- a limb support;
- a pivoting mechanism connected between the housing and the limb support for, in use, pivotably moving the limb support from a retracted position substantially parallel to the longitudinal axis of the housing to an extended position in which the limb support extends substantially perpendicularly to the longitudinal axis of the housing to support the lower limb.

The pivoting mechanism may comprise a pivot about which the first end of the limb support may pivot to move between the retracted position and the extended position.

The pivoting mechanism may be a linear actuator. The linear actuator may be battery-operated.

In another aspect of the present disclosure, there is provided a device for supporting a lower limb, comprising:

- a housing for a lower limb, the housing having a longitudinal axis aligned with a longitudinal axis of the lower limb, in use;
- a limb support;
- a driving mechanism connected between the housing and the limb support, the driving mechanism comprising a first part attached to the housing and a second part connected to the limb support;
- wherein, in use, movement of the first part of the driving mechanism in a direction parallel to the longitudinal axis of the housing drives movement of the limb support between a retracted position parallel to the longitudinal axis of the housing and an extended position in which the limb support extends perpendicularly to the longitudinal axis of the housing to support the lower limb.

The housing may include a leg portion to house a leg of a subject.

The housing may include a foot portion to house a foot of a subject.

The foot portion may have a first side and an opposed second side.

The foot portion may have a convex bottom surface. The convex bottom surface may comprise gripping material.

The foot portion may include a stabilizing portion extending from the bottom surface. The stabilizing portion may be configured to keep the device upright when the user is not wearing the device.

The stabilizing portion may be wedge shaped. The stabilizing portion may be configured to be compressible underfoot when the user is wearing the device.

The foot portion may include shock absorbent material. The shock absorbent material may be a relatively dense sponge.

The housing may include at least two rigid arms, each arm attached to each side of the foot portion to support the limb, in use.

The housing may include adjustable straps to secure a lower leg of the subject within the housing.

The housing may include a supporting cover that covers a substantial part of the back of the lower leg of the subject, in use.

The supporting cover may be attached to each of the at least two rigid arms.

The supporting cover may be adjustable to accommodate for different sizes of lower limb of a subject.

The limb support may comprise an elongated brace, the elongated brace may extend from a first end to a second end.

The limb support may comprise a base pivotally connected to the elongated brace.

The limb support may include an elongated brace and a base. The base may be pivotably connected to the elongated brace.

The limb support may further comprise a base lock to lock the base in a position substantially perpendicular to the elongate brace when the limb support is in the extended position.

The device may further include a fastener connected between the limb support and the housing to fasten the limb support to the housing when the limb support is in the retracted position. The fastener may be a release buckle.

The base may include two feet. Each of the two feet may have a convex lower surface. The convex lower surface may comprise a gripping material. First ends of each of the two feet may be angled towards each other. Second, opposing end of each of the two feet may be angled away from each other.

The driving mechanism may include a pivoting arrangement. The driving mechanism may be a mechanical linkage. The driving mechanism may have a first part and a second part.

The driving mechanism may comprise a handle for a user to grip and move the first part of the driving mechanism in a direction parallel to the longitudinal axis of the housing.

The handle may be faceted or curved.

The handle may be located adjacent the top of the first part of the driving mechanism.

The handle may be detachably attached to the first part of the driving mechanism via a fastener. The fastener may be an over centre fastener.

The first part of the driving mechanism may comprise a first elongate member. The first elongate member may extend from a first end to a second end.

The second part of the driving mechanism may comprise a second elongate member extending from a first end to a second end. The second end of the first elongate member may be pivotably connected to the first end of the second, substantially elongate member.

The second elongate member may be fixedly connected to the housing. The housing may include a restraining plate attached to the side of the foot via which the second substantially elongate member may be fixedly connected to the housing at an attachment point adjacent the first end of the second elongate member.

The elongated brace may comprise a guide extending along at least part of a length of the elongated brace. The guide may include a cavity. The guide may include a longitudinal slot contiguous with the cavity.

The second part of the driving mechanism may include a roller attached to the second end of the second elongate member. The roller may be located within the cavity of the elongated brace and moveable along the guide of the elongated brace. The roller may be a polyurethane roller.

The device may further comprise a driving mechanism lock to prevent movement of the first part of the driving mechanism in the longitudinal direction of the leg portion of the housing when the limb support is in the extended position or in the retracted position.

The driving mechanism lock may comprise a box longitudinally extending along an inner surface of the first elongated member. The box may have a longitudinally extending internal cavity. The box may have a slot contiguous with the internal cavity.

A roller may be attached to an arm of the housing adjacent the box. The roller may be located inside the cavity. Movement of the first elongated member parallel to the arm, moves the roller within the cavity along the slot. The roller may be a polyurethane roller.

The driving mechanism lock may include a window located through one side of the box. The driving mechanism lock may also include a stopper that can be moved into the window. Moving the stopper into the window when the roller is located between the window and the second end of the cavity may prevent movement of the roller within the cavity.

When the driving mechanism lock is disengaged, when the handle is pulled by a user in the longitudinal direction of the leg portion of the housing, pivotable movement of the first part relative to the second part may be caused. This movement, in turn, may cause movement of the second end of the second elongate member along the guide.

The arms of the housing, limb support and driving mechanism may substantially comprise a lightweight metal. The lightweight material may be aluminium.

Other aspects of the invention are also disclosed.

BRIEF DESCRIPTION OF THE DRAWINGS

Notwithstanding any other forms which may fall within the scope of the present invention, embodiments of the invention will now be described, by way of example only, with reference to the accompanying drawings in which:

FIG. **1** illustrates a device for supporting a lower limb in accordance with an embodiment of the present invention, when the limb support is in the retracted position, in situ; and

FIG. **2** is a side view of the device for supporting a lower limb in accordance with the embodiment of the present invention shown in FIG. **1**; and

FIG. **3** illustrates the device for supporting a lower limb in accordance with an embodiment of the present invention shown in FIG. **1**, when the limb support is in the extended position, in situ; and

FIG. **4** is a side view of the device for supporting a lower limb in accordance with an embodiment of the present invention shown in FIG. **1** when the limb support is in the extended position.

FIG. **5** is a back view of the device for supporting a lower limb in accordance with an embodiment of the present invention shown in FIG. **1**; and

FIG. **6** is a view of the underside of the housing of the device for supporting a lower limb in accordance with an embodiment of the present invention shown in FIG. **1**; and

FIG. **7** is a perspective view of the device for supporting a lower limb in accordance with an embodiment of the present invention shown in FIG. **1**; and

FIG. **8** is a partial view of base lock for supporting a lower limb in accordance with an embodiment of the present invention shown in FIG. **1**; and

FIG. **9** is another partial view of the base lock for supporting a lower limb in accordance with an embodiment of the present invention shown in FIG. **1**; and

FIG. **10** is view of the top end of the device in accordance with an embodiment of the present invention shown in FIG. **1**; and

FIG. **11** is view of the driving mechanism lock in accordance with an embodiment of the present invention shown in FIG. **1**.

DESCRIPTION OF EMBODIMENTS

It should be noted in the following description that like or the same reference numerals in different embodiments denote the same or similar features.

A device for supporting a lower limb according to a first aspect of the invention is generally indicated by the numeral **1000**. When a user of the device is seated on a bench or a chair, for example, and is wearing the device **1000** (as shown in FIG. **1**), the device **1000** can be used to elevate the lower leg above the ground and also, to support the lower limb at a position above the ground when the knee is extended.

The device **1000** comprises a housing **100** for the lower limb. The housing **100** is substantially L-shaped and comprises a vertical leg portion **110** and a horizontal foot portion **120** which together also define an L-shaped cavity **130** configured to house the limb. The leg portion **110** has a longitudinal axis A which is substantially parallel to the longitudinal axis of the lower leg within the housing **100** when the device **1000** is worn by the user.

The device **1000** comprises a limb support **200** which is moveable between a retracted position (shown in FIG. **2**) in which it is substantially parallel to the longitudinal axis of the leg portion of the housing **100**, to an extended position (shown in FIG. **3**) in which the limb support **200** extends substantially perpendicularly to the longitudinal axis of the leg portion of the housing **110**.

The device **1000** comprises a pivoting arrangement between the housing **100** and the limb support **200** via which the limb support **200** is moveable between the retracted position and the extended position.

In an embodiment (not shown), the pivoting arrangement may include a pin or shaft attached to the housing and extending through an end of the limb support to allow the limb support to rotate relative to the housing and move between the retracted position and the extended position. It is envisaged that a number of different types pivoting arrangements may be used to move the limb support from the retracted to the extended position, in other embodiments.

In the illustrated embodiment, the device **1000** includes a driving mechanism **500** connected between the housing **100** and the limb support **200**, which drives movement of the limb support **200** between the retracted position and the extended position. In the illustrated embodiments, this movement can be caused by force applied by the user to a part of the driving mechanism. In another embodiment (not shown), this movement can be caused by a motor or other type of driver.

In the illustrated embodiment, the driving mechanism **500** is a type of mechanical linkage. In an alternative embodiment, the driving mechanism **500** can include for example, a linear actuator or other driver. It is envisaged that a number of different driving mechanisms can be used in other embodiments.

FIG. **2** shows a seated user wearing the device **1000** when the limb support **200** is in the retracted position. FIG. **4** shows the user wearing the device **1000** when the limb support **200** is in the extended position.

The housing **100** is substantially L-shaped to accommodate the natural shape of a foot. As mentioned above, the housing **100** has a vertical component or a leg portion **110**. The housing **100** also has a horizontal component or a foot portion **120**. In this embodiment, the housing **100** is configured to house the lower leg of the patient while the patient wears an orthotic boot. In other embodiments, the housing **100** can be configured in size and shape to accommodate the lower limb of the patient while they are wearing normal shoes, or a plaster cast surrounding their lower limb,

The foot portion **120** has a first side **121**A and an opposed second side **121**B. The foot portion **120** is configured to house the foot portion of an orthotic boot including the foot of the user of the device or subject.

The foot portion **120** is substantially rectangular. Each side of the foot portion **121**A, **121**B has a metal plate **122**A, **122**B to which various parts of the housing **100** are connected.

The leg portion **110** extends from the foot portion **120** to a position under the knee so when the patient is seated, the device **1000** is located below the knee as can be seen in FIG. **1**.

As shown in FIG. **2**, the leg portion **110** includes a first arm **111**A extending vertically along the leg portion **120** from the first side of the foot portion **121**A. The leg portion also includes a second arm **111**B extending vertically along the leg portion **110** from the second side of the foot portion **120**. The first arm **111**A is opposed to the second arm **111**B. An end of each arm is also fixed to the foot portion **120**. Each of the first arm **111**A and second arm **111**B is sufficiently long to support the lower leg but does not extend pass the knee.

FIG. **5** shows a back view of the device **1000** when the limb support **200** is in the retracted position. The leg portion **110** also includes a third arm **113** fixed to the back of the foot portion and extending vertically along the back of the housing to support the back of the lower leg of the user, in use. There is a plurality of horizontally extending braces **115**, each of which are fixed to the first arm **111**A, the second arm **111**B and the third arm **113**. They also connect the first arm **111**A to the second arm **111**B and located adjacent the rear of the foot portion **124**. The braces mechanically support each of the first, second and third arms **111**A, **111**B, **113**. In this embodiment, there are three braces, as shown in FIG. **5**.

The third arm **113** extends partially up the back of the leg portion to a free end of the third arm **114**. Advantageously, the third arm **113** is configured such that the free end of the third arm **114** can flex outwardly to accommodate a larger calf of a subject. In the illustrated embodiment, each arm is relatively flat and slender. In this embodiment, each arm is made of aluminium. Advantageously, aluminium is lightweight. In other embodiments, the arms may be made of the other materials.

Each of the leg portion **110** and the foot portion **120** of the housing comprises a plurality of adjustable straps **105** configured to secure the lower limb within the cavity of the housing. In the illustrated embodiment, each of the foot portion **120** and leg portion **110** have two straps spaced from each other. In the leg portion **110** of the housing, the straps extend circumferentially around the leg portion **110**, over each of the three arms and are also secured to each of the three arms via screws. In the foot portion **120**, each strap is connected across the foot portion to secure the foot of a user (including an orthotic boot) within the housing **100**.

As can be seen in FIG. **2**, for example, each end of each strap across the foot portion **120** is attached onto the metal plates **122**A, **122**B on either of the two sides of the foot portion **121**A, **121**B with screws. Each strap is configured such that it is adjustable via a hook and loop fastener and an adjustment loop.

In use, the user will typically be wearing an orthotic boot to protect and support the injured lower leg. Advantageously, the straps **105** are adjustable to accommodate a variety of different sizes of orthotic boots worn by user.

The housing further includes a supporting cover **140**. The supporting cover **140** is configured to provide support to the limb within the leg portion of the housing and extends from the rear of the foot portion of the housing **124** to adjacent the underside of the knee. The supporting cover **140** is configured to at least partially cover substantially the whole length of the back of the leg. The supporting cover **140** is located internal to the three arms. The supporting cover **140** is fixed to the back of the foot portion **120** using screws.

In other embodiments, the supporting cover can be printed with a pattern or with drawings and be coloured in different colours. This may assist in the psycho-emotional aspect of recovery, especially when it

The supporting cover **140** has two wings **141**A, **141**B that extend from the back of the supporting cover to wrap partially around either side of the lower leg of the subject, in use. Each wing **141**A, **141**B is located near the top of the leg portion adjacent a calf of a lower leg, when the device is worn by a user. Each wing **141**A, **141**B also comprises a slot **142**A, **142**B. Each of the first and second arms **111**A, **111**B has an internally facing screw which engages with each respective slot of the first and second wings **142**A, **142**B. The diameter of each screw head is larger than the width of each slot to retain the supporting cover. The supporting cover **140** is moveable relative to each of the first and the second arms **111**A, **111**B within the constraint of the length of each slot **142**A, **142**B to accommodate for different sizes of lower limb so that the patient can comfortably wear the device **1000**.

The supporting cover **140** may be made of a sheet of tough plastic or fabric which is flexible but can substantially retain its shape while providing support to the lower limb, in use. User can

The foot portion, in use, is configured to support the driving mechanism and the leg portion of the housing as well as the lower limb of the user.

In this embodiment, the foot portion of the housing **120** includes a core made of plywood. Advantageously plywood is a lightweight material. In other embodiments, the core may be made of other lightweight material. Metal angle brackets fix ends of each of the first, second and third arms **111**A, **111**B, **113** to the core using screws. Each of the metal side plates of the foot **122**A and **122**B are also fixed to the core using screws.

The foot portion **120** also includes padding of a dense foam. In particular, the plywood is covered with or encased within a dense foam. The padding acts to reduce any impact on the bones of the injured limb during load bearing by cushioning the foot of the user, in use. In other embodiments, the food portion may have more padding for example where the user is not wearing a cam boot and is bare foot for example. In yet other embodiments, where there is already sufficient support provided by an orthotic boot, less padding may be needed in the foot portion of the housing **120**.

The foot portion **120** includes a padded sole **125** which will be in contact with the underside of the orthotic boot worn by the user, in use.

As can be seen in FIGS. **2**, **4** and **7**, the underside of the foot portion **126** has a convex curvature. This prevents the instance of any force concentrations in areas of the injured foot, particularly when the user is wearing the device **1000** when the limb support **200** is in the retracted position while the user is seated. This also prevents jarring of the bones of the foot or lower leg when the underside of the foot portion **1000** contacts the ground when the device is worn by the patient. This will reduce the chance of the user experiencing pain while wearing and walking with the device **1000**.

As mentioned above, the plywood is encased in relatively dense sponge. The dense sponge located between the plywood and strips is sufficiently shock absorbent to cushion the heel during heel-strike and improve comfort for the user. It is envisaged that in other embodiments, other types of shock-absorbent material can be used.

The underside of the foot portion also includes two grip strips **127**A, **127**B each extending longitudinally along the underside of the foot portion. In the illustrated embodiments one strip **127**A is located adjacent the first side of the foot portion **121**A and the other strip **127**B is located adjacent the second side of the foot portion **121**B. In this embodiment, the grip strips **127**A, **127**B are made of rubber and comprise a gripping pattern (not shown) to better engage with the ground to prevent the user slipping while wearing the device.

The foot portion **120** also includes a stabilising portion **128** which in the illustrated embodiments has a wedge shape and projects outwardly from the underside of the foot portion **120**. The stabilising wedge **128** acts to prevent the device **1000** from tipping and falling over due to gravity when it is placed on a flat surface in an upright position such as on the ground. Hence, as the underside of the foot portion **126** is curved, the greatest height of the wedge is located near the rear edge of the foot portion. This can be seen in FIG. **2** for example.

A resiliently flexible member (not shown) attaches the stabilising wedge **128** to the underside of the plywood. In the illustrated embodiment, the resiliently flexible member is a simple spring.

The limb support **200** includes a base **220**. In the illustrated embodiment, the base of the limb support **220** is relatively heavy and in a retracted position of the limb support **200**, is located near the top of the device **1000**. The stabilising portion **128** and spring are together configured to counterbalance the force of gravity on the top of the device that can act to tip the device over.

The stabilizing wedge **128** is made of a relatively soft sponge that is compressible underfoot such as when the user is walking while wearing the device. The spring is configured such that when the device **1000** is not loaded by the user, it is in its initial, extended position. Therefore, the spring is sufficiently resilient to remain in its extended position when the device **1000** is in an upright and retracted configuration. The spring is sufficiently flexible such that when the device is loaded by the user wearing the device **1000** and walking with the device **1000**, the spring and stabilizing wedge **128** compress underfoot.

Thus, the stabilising wedge **128** makes it easier for the user to grip and put on the device **1000** without the inconvenience of the device falling over when the holder is not holding the device **1000**. When the user is seated, it might be difficult to reach over and grab the device **1000** if it has fallen down. Especially, as the user is required to be careful when moving so as not to unnecessarily load the injured limb as this may cause pain. In the illustrated embodiments, the stabilising wedge **128** is located at the rear of the underside of the foot portion between the two grip strips **126**.

The convex shape of the underside of the device **126**, is configured for gradual load transfer when the user walks while wearing the device. During normal gait, the rear of the foot portion **124** will typically be loaded first and then load will be transferred along the length of the foot portion **120**. The stabilising wedge **128** is made of relatively soft sponge to cushion the foot of a patient. The sponge is sufficiently thick and otherwise configured such that there is no gap between the sponge and surrounding outer surface of the foot portion to prevent small rocks and other sediment or soil, for example, getting stuck between the sponge and the plywood.

In other embodiments, the housing **120** can comprise padding to make the device more comfortable to wear. This will enable the user to use the device while wearing a thinner or less robust orthotic boot. In yet another embodiment, the user may be able to use the device without wearing an orthotic boot. In this case, the orthotic boot may be incorporated within the cavity of the housing to support the injured lower leg.

As mentioned above, the limb support **200** is moveable from the retracted position to the extended position, and vice versa.

The limb support **200** comprises an elongated brace **210** to support the lower limb above the ground in an extended position of the lower limb, in use. As shown for example, in FIGS. **1** and **2**, the elongated brace **210** is located external to and at the rear of the housing **100** when the limb support **200** is in the retracted position.

The elongated brace **210** is sufficiently strong and stable to support the lower limb when the knee is extended.

The total height of the limb support **20** is selected to ensure that when the limb support is in the extended position, that the lower limb is substantially perpendicular to the limb support **200** when the device **1000** is worn. If the limb support **200** is too high then there may be unnecessary forces applied to parts of the lower limb such as the hip which may make the user uncomfortable.

In another embodiment (not shown), the limb support may be at a height at which the lower limb is not perpendicular to the limb support in use while being comfortable for the user.

In another embodiment (not shown), the limb support may have an adjustable height. It is envisaged that the skilled person will be able to make the height of the limb support adjustable in a number of different ways.

The elongated brace **210** comprises two slender arms **211**A, **211**B spaced horizontally from each other and located on either side of and at the rear of the housing **100** when the limb support **200** is in the retracted position. In the retracted position, each arm of the elongated brace **211**A, **211**B extends vertically from the rear of the foot portion of the housing **124**. Each arm **211**A, **211**B is pivotably connected to the rear of the foot portion of the housing **124**. Each arm has a first end **212**A, **212**B and a second end **213**B, **213**B.

Each arm **211**A, **211**B also includes a guide **215**A, **215**B to guide the movement of the limb support **200** between the retracted and the extended position. Each guide **215**A, **215**B is a slot longitudinally extending along the length of each of the two arms of the elongated support **211**A, **211**B. Each arm **211**A, **211**B has an inwardly facing side facing the housing and an outwardly facing side facing away from the housing. In the illustrated embodiment, each slot is located on the outwardly facing side and extends from adjacent the base **250** or near the second end of each arm **211**B, **212**B to approximately halfway along the elongated support **210**.

Each second part of the driving mechanism **520**A engages with each respective slot (as explained in detail below). The length of each slot is configured to allow the second part of the driving mechanism **520**B to move relative to the slot and cause the limb support **200** to move between the retracted position and the extended position. Each slot restrains the second part of the driving mechanism **520**A and controllingly guides the movement of the limb support **200** into the extended position or the retracted position.

The second end of the each of the arms **213**A, **213**B are pivotably connected to the base **250** such that the elongated brace **210** can be moved from a reduced position in which the base **250** is substantially parallel to the elongate brace **210** to a position in which the base is perpendicular to the elongate brace **210** for supporting the lower limb in the extended position of the limb support **200**.

The device **1000** includes a base lock **400** to lock the base **250** in a position perpendicular to the elongated brace **210**.

Advantageously, the base **250** can be folded behind the housing **100** in the reduced position as shown in FIG. **2** so as not to obstruct the user while the user is moving. This also results in a relatively compact and sleek device.

As can be seen in FIG. **6**, the base **250** also includes a pair of curved feet **251**A, **251**B, adjacent each other and spaced horizontally from each other. Each foot **251**A, **251**B has a convex outer surface **252**A, **252**B for contacting the ground that is covered with a gripping material. The gripping material comprises a tread (not shown) for engaging with the ground when the outer surface of each of the feet **252**A, **252**B are in contact with the ground when the limb support **200** is in the extended position.

As shown, for example, in FIG. **5**, when the limb support **200** is in the retracted position and the base **250** is in a position parallel to the limb support **200**, the upper ends of the feet **257**A, **257**B are turned outwardly while the lower ends of the feet **258**A, **258**B are angled inwardly. Advantageously, this is so that the upper ends of the feet **257**A, **258**B do not protrude into the thigh of the subject when the subject is wearing the device and seated, and the limb support **200** is in the retracted position.

Another advantage of this is that when the limb support **200** is in the extended position, the outwardly angled nature of the feet **251**A, **251**B reduces the likelihood of the base **250** from slipping backwards or forwards when the subject moves compared to if the feet were parallel to each other. There is greater surface area of contact between the sides of the feet and the ground when the feet are outwardly angled compared to when they are parallel to each other. This increases the friction between the ground and each of the feet.

A flange **253**A, **253**A extends perpendicularly from approximately the centre of each of the feet **257**A, **257**B. As shown in FIG. **6**, the two curved feet are connected to each other by a bridge **255** fixed to each of and extending between the two flanges **253**, **254**.

The bridge **255** has a substantially rectangular portion extending between the two flanges. The bridge **255** also has two end portions **256**A, **256**B extending substantially perpendicularly from either side of the rectangular portion. Each of the two end portions **256**A, **256**B is pivotably connected to a respective second end of a respective arm of the elongated brace **213**A, **213**B with a suitable fastener. It is envisaged that a number of known fasteners can be used to pivotably attach each end portion **256**A, **256**B to the respective second end of an arm of the elongated brace **213**A, **213**B.

As mentioned above, the limb support **200** comprises a base lock **400** to lock the base in a position substantially perpendicular to the elongate brace when the limb support is in the extended position. In use, when the base lock **400** is disengaged, the user can grip the bridge **255** to rotate and move the base about the second end of the elongated support.

The base lock **400** has two identical parts **400**A, **400**B located internal to each end portion of the bridge **256**A, **256**B. Each part of the base lock comprises a lever **410**A, **410**B each having a first end **411**A, **411**B, and a second end **412**A, **412**B and a resiliently flexible member **450**A, **450**B.

FIGS. **8** and **9** depict one side of the base lock **400**A, the resiliently flexible member **450**A is located between and fixed to each of the lever **410**A and the respective end portion of the bridge **256**A. The resiliently flexible member **450**A provides a fulcrum **452**A about which the lever pivots to move a second end of the lever **412**A. Applying a force at a first end of the lever **411**A to move the first end of the lever towards the adjacent perpendicular portion of the bridge **256**A, moves the second end of the lever **412**A away from the second end of the respective arm of the elongated member **213**A. This is illustrated in FIG. **8**.

The lever **410**A can be made of a suitable metal such as aluminium and steel and is rigid. The lever **410** is shaped to follow part of the shape of the inner surface of the bridge as shown in FIG. **9**, so as not to protrude into the cavity of the housing **130**, in use.

The first end of the lever **411**A is sufficiently wide to allow a user to comfortably push the first end of the lever **411**A down with their thumb.

Each part of the base lock **400**A also includes two screws **413**A extending through and fixed to the second end of the lever **412**A. As shown in FIG. **9**, the bodies of the screws extend in a direction towards the second end of the respective arm of the elongated member **213**A.

When the base **450** is in the perpendicular position, each screw **413**A is aligned with a respective one of two holes **414**A extending through the second end of the respective arm of the elongated member **213**A.

Each hole **414**A is configured to receive and retain a substantial portion of the bodies of each screw to prevent the base **250** moving out of the perpendicular position.

The resiliently flexible member **450** is attached to the lever **410**A at a position approximately midway between the first end **411**A and the second end of the lever **412**A. The resiliently flexible member **450**A is a thin sheet of metal bent to define three planar parts. A first planar part **451**A is fixed to the perpendicular portion and an opposed third planar part **453**A is fixed to the lever and a second planar part extending between the first and the third parts. There is bend between the second and the third parts. The second bend acts as the fulcrum **452**A about which the lever **410**A pivots.

The base lock **400**A also includes a restraint **430**A which locates the lever **410**A relative to the inner surface of the bridge **255**, prevents lateral movement of the lever **410**A. The restraint **430**A also acts to control the movement of the second end of the lever **412**A when the first end of the lever **411**A is depressed by the thumb of the user.

The restraint **430**A comprises a screw having a body **431**A extending from a head of the screw **432**A located external to an outer surface of the respective end portion of the bridge **256**A, through the respective end portion **256**A, between the respective end portion **256**A and the lever **410**A and through the lever **410**A to an outer surface of the lever **410**A. The end of the body of the screw is fixed to the lever **410**A such that the lever **410**A cannot move along the body of the screw **431**A.

The head of the screw **432**A is spaced from the outer surface of the respective end portion by a helical spring **436**A around the part of the body of the screw that is external to the respective end portion of the bridge **256**A.

The spring **436**A is biased to prevent any movement of the head **432**A towards the end portion and ensure that the two screws of the second end of the lever **412**A are retained within the holes in the second end of the elongated member **414**A, in the absence of any user-applied forces on the first end of the lever **411**A.

To unlock the base **250** from an engaged position of the lock **400**, as shown in FIG. **9** the user can simultaneously press on both levers **410**A, **410**B until the screws are moved out of the holes as shown in FIG. **9**. This movement releases the base **250** from the second end of the respective arm of the elongated brace **213**A and the user can move the base **250** into the position parallel to the elongated member.

In other embodiments, it is envisaged that other types of locks can be used to lock the base to the elongated brace.

As mentioned above, the driving mechanism **500** drives the movement of the limb support **200** from the retracted position to the extended position. In the illustrated embodiment, the driving mechanism **500** is a type of mechanical linkage. In another embodiment (not shown), the driving mechanism **500** can include a battery-operated linear actuator that is actuated by a button. It is envisaged that a number of different driving mechanisms can be used to drive the limb support **200** between the retracted position and the extended position. The driving mechanism **500** includes a pivot about which the limb support **200** rotates relative to the housing **100**.

The driving mechanism **500** as shown in the illustrated embodiments extends between the limb support **200** and the arms **111**A, **111**B of the leg portion of the housing on either side of the device. The driving mechanism and has a first side **500**A and an identical second side **500**B.

The first side of the driving mechanism **500**A includes a first part **510**A and a second part **520**B and is connected to a first arm of the elongated support **211**A. The second side of the driving mechanism **500**B also includes a first part **520**A and a second part **520**B and is connected to the second arm of the elongated brace **211**B. In this way, each side **500**A, **500**B of the driving mechanism **500** separately and simultaneously drives the respective side of the limb support that it is connected to.

The first part of the driving mechanism **510**A is a first elongate member extending from a first end **511**A to a second end **512**A. The second part of the driving mechanism **520**A comprises a second substantially elongate member which also extends from a first end **521**A to a second end **522**A. The second end of the first elongate member **512**A is pivotably connected to the first end of the second elongate member **521**A.

There is a polyurethane roller (not shown) attached to and extending from the surface of the second end of the second elongate member **522**A that is facing the slot of the guide **215**A. The arm of the elongated support includes a longitudinally extending internal cavity and a slot **215**A contiguous with the internal cavity.

The polyurethane roller is seated within the internal cavity of the elongated support of the limb support and is configured to move along the length of the slot **215**A and therefore, allow the limb support to move between the retracted and extended positions.

As mentioned previously, the driving mechanism **500** is configured such that pivotable movement of the first part **510**A relative to the second part **520**A causes movement of the second end of the second elongate member along the guide.

The driving mechanism **500** also comprises a handle **550** attached between and connecting the two sides of the driving mechanism. In particular, the first part of the driving mechanism comprises a handle **550** for a user to grip and move the first part of the driving mechanism in a direction parallel to the longitudinal axis of the leg portion of the housing **110**.

As mentioned previously, movement of the first part of the driving mechanism **510**A in a direction parallel to the longitudinal axis of the leg portion of the housing **100** drives movement of the limb support **200** between a retracted position parallel to the longitudinal axis of the housing and an extended position in which the limb support extends perpendicularly to the longitudinal axis of the housing to support the lower limb.

The handle **550** also extends from a first end **551**A to a second end **551**B. The handle **500** projects outwardly from the housing **100**. The handle **550** is also faceted such that there is enough space between the handle **550** and the lower limb for the user to insert their fingers under the handle **550**, grip and move the handle **550** to drive movement of the limb support **200**. In another embodiment (not shown), the handle **550** can be curved with a convex inner surface facing the limb of a user, in use.

The handle **550** is also detachable from the first part **510**A of the driving mechanism.

The handle **550** has an outer surface and an inner surface. FIG. **2** shows each end of the handle **551**A, **551**B is engaged with and attached to the respective first part of each side the driving mechanism **510**A, **510**B via a respective fastener **560**A, **560**B. In this embodiment, the fastener **560**A, **560**B is a type of over-centre fastener with a catch plate or "mousetrap" fastener.

FIG. **10** illustrates the fastener **560**B. The catch plate **561**B of the fastener **560**B includes a curved retainer configured to engage with and retainingly hold a loop **563**B attached to a body of over-centre fastener **562**B. The catch plate **561**B is fixed to the outer surface of the handle and adjacent the end of the handle **551**B.

The body of the over-centre fastener **562**B is attached to an outer side of the top of the second arm of the driving mechanism **511**B and includes a loop **563**B to engage with the retainer.

The handle **550** and the top end of each side of the first elongate member of the driving mechanism **511**B, **511**A together also include an alignment guide **590**A, **590**B to correctly align each catch plate **561**A, **561**B with the body of the over centre fastener **562**A, **562**B, such that each loop of the body of the over centre fastener **563**B, **563**A can easily engage with the catch plate and the end of the handle **551**A, **511**B and top end of the first part provide a smooth surface adjacent the limb, in use, when the handle is connected to the first part of the driving mechanism.

The alignment guide **590**A has a female part or recess **591**A located under the catch plate. The alignment guide **590**A also has a male part **592**A configured to receive the female part **591**A when the first and second parts of the alignment guide are brought together. The male part **592**A is located under the body of over-centre fastener **562**A, **562**B.

The second substantially elongate member **520**A, **520**B of the driving mechanism **500**A, **500**B has a first portion **523**A, **523**B that extends out of longitudinal alignment with the second portion **524**A, **524**B. There is a bend defined between the two portions **525**A. It is envisaged that in other embodiments, the first portion will be in longitudinal alignment with the second portion i.e. the second substantially elongate member **520**A, **520**B will be a single elongate member.

As depicted in FIGS. **3** and **4**, the driving mechanism **500**A also includes a restraining plate **540**A secured to and extending from the side of the foot portion **121**A of the housing. The second elongate member **520**A is pivotably connected to the free end of the plate at the bend **525**A between the first **523**A and second **524**A portions. The restraining plate **540**A mechanically supports each of the first and second elongate members when the limb support **200** is in the extended position and prevents unnecessary lateral movement of the driving mechanism **500**A while the limb support **200** is moved from the retracted to extended positions.

The relative dimensions of the first and second elongate members of the driving mechanism, the position of the plate along each side of the foot portion and slot length of the limb support have been selected such that the second end of the limb support fits closely to the housing in the retracted position of the limb support and that the limb support extends substantially perpendicularly to the housing to support the lower limb in the extended position.

Each side of the limb support **200**A, **200**B is releasably connectable to the adjacent respective side of the first part of the driving mechanism **510**A, **510**B via a fastener **280**A, **280**B. As shown in FIG. **6**, the fastener **280**A is a side release buckle including a receiving portion **281**A and an engaging portion **282**A. The engaging portion **282**A for each buckle is attached to the limb support **200**A to an end of the end portion of the bridge **256**A. The receiving portion **281**A for each buckle is attached to the first part of the driving mechanism under and adjacent the handle **550**. This location of the fastener **280**A makes it easy for the user to reach and unbuckle the fastener **280**A while wearing the device **1000**.

It is envisaged that a variety of different types of fasteners can be used to secure the limb support to the first part of the driving mechanism in the retracted position of the limb support.

Each part of the driving mechanism **500**A, **500**B also includes a driving mechanism lock **570**A, **570**B to prevent movement of the first part of the driving mechanism **510**A, **510**B relative to the housing **100** to ensure that the limb support **200** is locked in the extended position.

As can be seen for example, in FIG. **5**, each side of the driving mechanism **500**A, **500**B is configured such that the first elongate member **510**A, **510**B is parallel to and adjacent the arms of the leg portion of the housing **111**A, **111**B.

The driving mechanism lock **570**B includes a longitudinally extending box **571**B with an internal cavity (not shown) and a longitudinally extending slot **573**B that is contiguous with the internal cavity **572**B. The box **571**B is fixed to an inner surface of the first elongate member **510**B with the slot **573**B facing the inner surface of the first elongated member.

It is envisaged that in another embodiment (not shown), the first elongated member may include the internal cavity and the longitudinally extending slot. In this embodiment (not shown), a separate box is not required.

There is a polyurethane roller **574**B attached to an outer surface at the top end of the arm **111**B. The polyurethane roller **574**B is located within and moveable along the internal cavity along the longitudinally extending slot **573**B. The slot **573**B is sized that the roller **574**B is retained within the internal cavity.

The driving mechanism lock **570**B further comprises a removeable stopper **575**B attached to the outer side of first elongate member **510**B of the driving mechanism. There is a window **576**B located at and extending through a front facing surface of the box. One end of the cavity is located adjacent the handle and the other end of the cavity is located at a point along the length of the first member **579**A. The window **576**A is located above the other end of the internal cavity.

The stopper **575**B is made of metal bent into a U-shape. Therefore, the stopper has a first end **577**B and a second end **578**B separated by a bend. A first moveable member **579**B is attached to and pivotably moveable relative to the external surface of the first elongate member of the driving mechanism **510**A. The first moveable member has a first end **580**B and a second end **581**B. A second moveable member **582**A has a first end **583**B, and a second end **584**B fixed to the external surface of the first elongate member of the driving mechanism **510**B.

The second end of the stopper **578**B is located between and fixed to each of the second end of the first moveable member **581**B and the first end of the second moveable member **579**B as shown in FIG. **11**.

The first moveable member **577**B has an inflection **582**B adjacent the second end of the first moveable member **581**B configured such that the first moveable member **577**B and the second moveable member **578**B can fit parallel to each other when aligned vertically i.e. that the first end **583**B can fit into the gap created by the inflection.

When the first part of the driving mechanism **510**B is moved parallel to the limb to move the limb support **200**B from a retracted to an extended position, the polyurethane roller **574**B moves within the internal cavity and along the slot **573**B. The box can be made of metal and sized to house the polyurethane roller. Advantageously, there is low friction between the polyurethane roller **574**B and the metal of the box which enables smooth movement of the roller **574**B within the cavity **572**B.

The length of the slot **573**B and cavity **572**B are each configured such that the polyurethane roller **574**B is at the top end of the longitudinal cavity **572**B in the retracted position of the limb support. The slot **573**B is also configured such that the polyurethane roller **574**B located between the window **576**B and the second end of the longitudinal cavity **575**B in the fully extended position of the limb support **200**.

Before the driving mechanism **500**B is engaged to move the limb support **200**B from the retracted to the extended position, the driving mechanism lock **570**B is disengaged as shown in FIG. **11**. When the limb support **200** is in the extended position, the user can push the unfixed end of the stopper **577**B into the window **576**B to prevent the polyurethane roller **574**B moving towards handle **550** within the cavity **572**B.

To engage the driving mechanism lock **570**B, the user pushes the first end of the first moveable member **580**B into vertical alignment with and into a position that is parallel to the outer surface of the first elongate member of the driving mechanism **510**B. This causes the second end of the first moveable member **581**B and the second moveable member **582**B to also move into parallel alignment with the first part of the driving mechanism **510**B. This also causes the first end of the stopper **577**B to enter the window above the stopper **576**B. In this position, the first end of the first moveable member **580**B abuts with a horizontal surface **585**B located under the end of the handle **551**B such that any movement of the handle **550** towards the foot of the device **120** is prevented by the lock **570**A.

The first end of the first moveable member **570**A also has an outer rubber sleeve **586**A to increase the friction between the horizontal surface **585**A and the first end **580**A so that the chance of the first end of the first moveable member **580**A slipping relative to the horizontal surface **585**A is reduced, for example, when the user pushes down on the handle **550** in a locked position of the driving mechanism **500**A, **500**B while the limb support **200** is in the extended position.

To disengage each driving mechanism lock **570**A, **570**B the user simply pushes the first end of the first moveable member **580**A, **580**B out of vertical alignment with or out of a position that is not parallel to the outer surface of the first part of the driving mechanism **510**A, **510**B. The second end of the first moveable member **581**A, **582**A moves away from the first part of the driving mechanism **510**A, **510**B and causes removal of the first end of the respective stopper **577**A, **577**B from the cavity through the respective window **576**A, **576**B.

The use of the device **1000** in accordance with the illustrated embodiment will now be described. Before the user wears the device **1000**, the limb support **200** is in the retracted position and the limb support **200** is fastened to the first part of the driving mechanism **510**A, **510**B via the respective fastener **570**A, **570**B. The driving mechanism lock **570**A, **570**B is also disengaged.

The handle **550** is removed by disengaging the over-centre fasteners **560**A, **560**B located on either side of the handle **550** so that the handle **550** does not obstruct inserting the lower limb into the housing **100**.

A user while seated on a chair or a bench for example, inserts their lower leg which is covered by an orthotic boot, foot first into the cavity of the housing device **110** while the limb support **200** is in the retracted position. The user can then secure their lower leg within the device **1000** by adjusting the straps until they feel their lower leg is secure within the device **1000**. The user can then reattach the handle **550** to the first part of the driving mechanism **510**A, **510**B by aligning the first part of the alignment guide **591**A, **591**B on each side of the handle **550** with the second part of the alignment guide **592**A, **592**B on the first end of the first part of the driving mechanism **510** such that the protrusions are inserted into the recess, and engaging each loop of the over centre fastener with the respective retainer of the catch plate **561**A.

The user can rest the leg while wearing the device **1000** on the ground. Once the lower leg is secured within the device **1000**, the user unbuckles the buckle fastener **280**A, **280**B to disconnect the limb support **200** from the first part of the driving mechanism **510**A, **510**B. The user then pushes the limb support **200** away from the housing **100**. The user then raises their lower limb and therefore, the device **1000** above the ground. This gives the user some space underneath their knee to manipulate the limb support **200**.

The user unlocks the base **250** by holding down the first ends of both levers **410**A, **410**B and simultaneously moving the base **250** from the parallel position to a position perpendicular to the elongated brace **211**A, **211**B by rotating the base **250** about the elongate support **211**A, **211**B. The user then locks the base **250** in the perpendicular position by releasing the levers **410**A, **410**B until the second ends of the levers engage with the holes in each respective second end of elongated brace.

The user then grips the handle of the driving mechanism **550** and pulls it parallel to the longitudinal axis of the housing in a proximal direction (towards themselves) while extending their knee. Upon proximal movement of the handle **550**, the first elongate member **110** translates along the lower limb and moves parallel to the arms of the housing, and the second elongate member simultaneously rotates approximately about the restraining plate. While the second elongate member rotates, the second end of the second elongate member translates along the slot from the first end of the slot to the second end of the slot. As the second elongate member is rigid, this movement of the second elongate member drives the elongate limb from the retracted position to the extended position. When the second end of the slot has traversed the entire length of the slot, the limb support is in the extended position.

In the extended position, the handle is at a position above the knee.

The driving mechanism lock is then engaged into the locked position to lock the limb support in position and to prevent movement of handle **550** in the opposition direction.

To return the limb support **200** to the retracted position, the driving mechanism lock **570**A, **570**B on either side of the device is disengaged and the user pushes on the handle. The second end of each second elongate member **522**A, **522**B moves from the second end of the slot back to the first end of the slot and pulls the limb support **200** towards the back of the housing. When the base **250** is sufficiently close, the user can readily grip the base **250**, unlock the base lock **400** and fold the base **250** down into the position parallel to the limb support **200**. The user can then connect the limb support **200** to the first part of the driving mechanism using each buckle **280**A, **280**B.

To remove the device, the user can remove the handle **550**, loosen the straps **105** and then, carefully pull the device **550** off the lower limb.

The advantage of the mechanical driving mechanism illustrated in the embodiments is that it is simple, easy to use, robust and long-wearing. The user can use the device for an extended period of time such as for a number of years without having to maintain or replace the device.

Further, the position of the handle, buckle, base, base lock and driving mechanism lock are selected to be within reach of the user when they are seated and wearing the device, so that the user can easily and comfortably wear and deploy the device.

Advantageously, the buckle, base lock and driving mechanism lock in the illustrated embodiment do not require much manual dexterity or force to use. Therefore, the user will be motivated to use the device and to continue using the device to elevate their lower leg.

The driving mechanism, limb support, arms of the housing and other parts of the device which require strength and stiffness such as the plates on each side of the foot portion can be made of a suitable metal such as steel or aluminium. Alternatively, a suitably strong and stiff plastic may be used. In another embodiment, a composite material can be used such as a glass-reinforced plastic.

In other embodiments, other parts of the device can also be made of plastics with suitable mechanical properties.

The device is advantageously lightweight so that it is easy to carry and so that it does not unnecessarily load the injured lower leg while worn. The device is configured such that the user can walk with the device on the lower limb.

The device can be made in a number of sizes to suit different age groups and demographics of users.

It is envisaged that the foot portion of the housing can be constructed in a number of different ways such as molding using a suitable material such as types of plastics.

The skilled person will appreciate that pivotable connections between parts can be provided in a number of different ways including nuts and bolts with bearings to allow for the desired movement.

It is envisaged that the above device can be used to provide therapy for a variety of medical issues associated with a lower limb which require elevation of the lower limb such as fractures in the ankle, foot and shin, ankle sprains, achilles tendon injuries or calf muscle tears.

As mentioned above, the housing of the device can be modified to accommodate for different types of orthotics around the injured lower leg while still being comfortable for the user. For example, the housing can be modified so that a diabetic patient can comfortably wear the device while wearing normal shoes. For example, the padding housing can also be modified for a person wearing an orthotic boot, or plaster or a sand shoe or other shoes such as an ugg boot, for example.

It is envisaged that other parts of the device can be modified without departing from the scope of the invention.

Interpretation

Embodiments

Reference throughout this specification to "one embodiment" or "an embodiment" means that a particular feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, appearances of the phrases "in one embodiment" or "in an embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment, but may do so. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner, as would be apparent to one of ordinary skill in the art from this disclosure, in one or more embodiments.

Similarly it should be appreciated that in the above description of example embodiments of the invention, various features of the invention are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of one or more of the various inventive aspects. This method of disclosure, however, is not to be interpreted as reflecting an intention that the claimed invention requires more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive aspects lie in less than all features of a single foregoing disclosed embodiment. Thus, the claims following the Detailed Description of Specific Embodiments are hereby expressly incorporated into this Detailed Description of Specific Embodiments, with each claim standing on its own as a separate embodiment of this invention.

Furthermore, while some embodiments described herein include some but not other features included in other embodiments, combinations of features of different embodiments are meant to be within the scope of the invention, and form different embodiments, as would be understood by those in the art. For example, in the following claims, any of the claimed embodiments can be used in any combination.

Comprising and Including

In the claims which follow and in the preceding description of the invention, except where the context requires otherwise due to express language or necessary implication, the word "comprise" or variations such as "comprises" or "comprising" are used in an inclusive sense, i.e. to specify the presence of the stated features but not to preclude the presence or addition of further features in various embodiments of the invention.

Any one of the terms: including or which includes or that includes as used herein is also an open term that also means including at least the elements/features that follow the term, but not excluding others. Thus, including is synonymous with and means comprising.

Scope of Invention

Thus, while there has been described what are believed to be the preferred embodiments of the invention, those skilled in the art will recognize that other and further modifications may be made thereto without departing from the spirit of the invention, and it is intended to claim all such changes and modifications as fall within the scope of the invention. For example, any formulas given above are merely representative of procedures that may be used. Functionality may be added or deleted from the block diagrams and operations may be interchanged among functional blocks. Steps may be added or deleted to methods described within the scope of the present invention.

Although the invention has been described with reference to specific examples, it will be appreciated by those skilled in the art that the invention may be embodied in many other forms.

INDUSTRIAL APPLICABILITY

It is apparent from the above, that the arrangements described are applicable to the medical device and health-care industries.

The invention claimed is:

1. A device for supporting a lower limb of a subject, comprising:
   - a housing for a lower limb, the housing having a longitudinal axis configured to be aligned with a longitudinal axis of the lower limb, in use;
   - a limb support comprising an elongated bracing member;
   - a driving mechanism connected between the housing and the limb support, the driving mechanism comprising a first part attached to the housing and a second part connected to the limb support; and
   - an actuator configured to be located at or near a knee of the subject wearing the device and connected to the driving mechanism for enabling the subject to operate the driving mechanism,
   - wherein, in use, actuation of the driving mechanism drives a movement of the limb support between a retracted position parallel to the longitudinal axis of the housing and an extended position in which the limb support extends perpendicularly to the longitudinal axis of the housing to support the lower limb, and
   - wherein in the extended position, the limb support is adapted to extend between the housing and the ground to support the lower limb above the ground.

2. The device of claim **1**, wherein the housing includes a foot portion configured to house a foot of the subject.

3. The device of claim **2**, wherein the housing includes at least two rigid arms, each rigid arm attached to a first and second side of the foot portion to support the lower limb, in use.

4. The device of claim **3**, wherein the housing includes a supporting cover attached to each of the two rigid arms that covers a substantial part of a back of the lower leg of the subject, in use.

5. The device of claim **4**, wherein the supporting cover is adjustable to accommodate for different sizes of lower limb of the subject.

6. The device of claim **2**, wherein the foot portion has a convex bottom surface comprising a gripping material.

7. The device of claim **2**, wherein the foot portion includes a stabilizing portion extending from the bottom surface, the stabilizing portion being configured to keep the device upright when the subject is not wearing the device.

8. The device of claim **1**, wherein the limb support comprises a base pivotally connected to the elongated bracing member.

9. The device of claim **8**, wherein the limb support further comprises a base lock to lock the base in a position substantially perpendicular to the elongated bracing member when the limb support is in the extended position.

10. The device of claim **8**, wherein the base has a convex lower surface.

11. The device of claim **1**, wherein the driving mechanism comprises the first part comprising a first elongate member, the first elongate member including a first end and a second end,
    - wherein the driving mechanism further comprises the second part comprising a second elongate member, the second elongate member including a first end and a second end, the second end of the first elongate member pivotably connected to the first end of the second elongate member, and
    - wherein the actuator is a handle fixed to the driving mechanism for the subject to grip to move the first part of the driving mechanism in a direction parallel to the longitudinal axis of the housing to drive the limb support between the retracted position and the extended position.

12. The device of claim **11**, wherein the handle is detachable.

13. The device of claim **11**, wherein the actuator drives the first elongate member which interacts with a slide to allow the first elongate member to be driven substantially along the housing, and
    - wherein the second end of the second elongate member slides in a captured manner in a linear recess in the elongated bracing member of the limb support to enable the limb support to stably move between the retracted position and the extended position.

14. The device of claim **1**, wherein the elongated bracing member comprises a guide extending along at least part of a length of the elongated bracing member, the guide being configured to engage with the second part of the driving mechanism.

15. The device of claim **14**, wherein the guide comprises a cavity, and
    - wherein the device further includes a roller attached to a second end of a second elongate member, the roller being located within and moveable along the cavity of the elongated bracing member, and moveable along the guide of the elongated bracing member.

16. The device of claim **1**, wherein the housing includes a leg portion configured to house a leg of the subject.

17. The device of claim **1**, wherein the housing includes adjustable straps configured to secure a lower leg of the subject within the housing.

18. The device of claim **1**, wherein the device further includes a fastener connected between the limb support and the housing to fasten the limb support to the housing when the limb support is in the retracted position.

19. The device of claim **1**, wherein the device further comprises a driving mechanism lock to prevent movement of the first part of the driving mechanism relative to the housing when the limb support is in the extended position or in the retracted position.

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